



MARIANO MARCOS STATE UNIVERSITY



BAGONG PILIPINAS

Cultivating Minds, Transforming Futures



Applied**HE**TM
PUBLIC UNIVERSITY RANKING: **ASEAN 2024**
RANK 2 AMONG PHILIPPINE SUCs



Rank 601-800 in the World
Rank 3 among Philippine SUCs

WURI | **Rank 101st-200th**
The WORLD UNIVERSITY
RANKINGS for INNOVATION

Sentinel

A 2D Simulation-Based Decision Support System for Flood Evacuation Planning

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Motivation

What Drives This Research?

The Problem

Flooding and typhoons can quickly make roads inaccessible and increase the number of people who need to evacuate.

Local responders need to determine:

- Which routes are still usable?
- Which evacuation center should be used?
- What happens if a road becomes flooded?
- What happens if an evacuation center becomes full?

Existing maps can show locations and hazard areas, but they provide limited ways to **test** different evacuation situations before they happen.

Sentinel aims to help responders prepare by allowing them to simulate possible flood scenarios and evaluate evacuation options.

Problem

1. Limited Scenario Testing

Existing mapping tools mainly show information but do not allow users to easily test different flood situations.

2. Changing Evacuation Routes

Flooded or blocked roads can make planned evacuation routes unusable.

3. Evacuation Center Capacity

The number of affected residents may exceed the capacity of nearby evacuation centers.

4. Resource Allocation

Responders need to determine how available vehicles and other resources can be distributed.

Research Gap

There is a need for a localized simulation tool that allows disaster planners to test "what-if" situations and compare possible evacuation decisions.

Proposed Solution

Our Research-Based Solution

Sentinel is a **2D interactive disaster simulation and decision-support system** designed for flood evacuation planning.

Users can create different scenarios by changing conditions such as:

- Flooded roads
- Blocked bridges
- Number of affected people
- Evacuation center capacity
- Available rescue resources

The system then simulates the scenario and provides:

- Recommended evacuation routes
- Evacuation center allocation
- Estimated evacuation time
- Resource allocation
- Scenario comparison

Proposed Solution: Sentinel



(proprietary tools, limited testing)

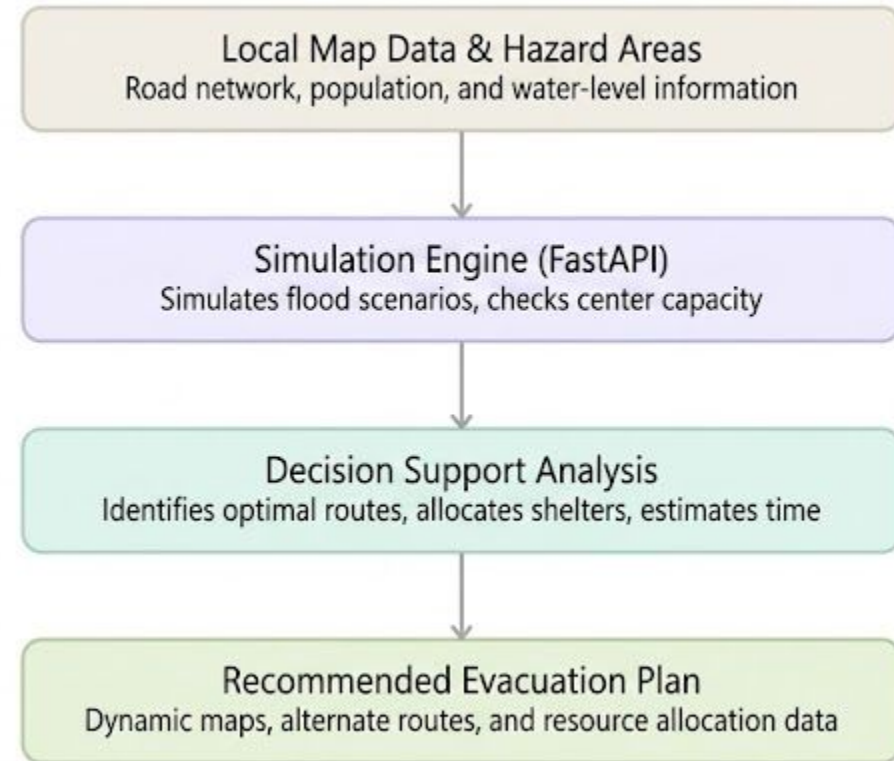


OpenStreetMap + Python - Local Data Integration

OPEN-SOURCE, DISASTER MANAGEMENT & OPTIMIZATION FRAMEWORK

(Sentinel solution, unlimited scenarios!)

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Research Thrust and SDG Alignment

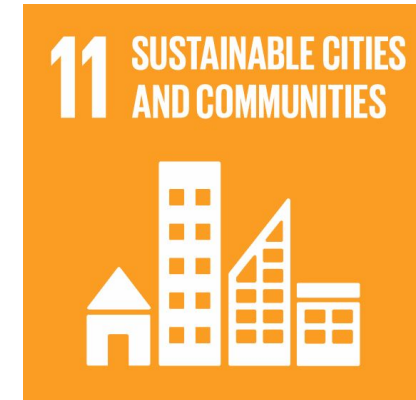
Alignment with Research Priorities and Sustainable Goals

Research Thrust



Scientific Computing and Optimization

Sustainable Development Goals (SDGs)





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